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LogDigitizer

USER GUIDE

Digitization of geophysical well logs

From scanned log image to calibrated digital curve

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1. Getting started

LogDigitizer converts scanned geophysical well logs into digital curves that can be exported in LAS 2.0 format. You prepare the image, calibrate its depth and curve scales, then trace the curves using semi-automatic tools and manual corrections.

This guide follows that workflow from a scanned image to an exported LAS file. Start with one image and one curve; the later chapters explain how to handle additional scales, wrap-around curves and multiple images.

1.1. Install and launch

Run the LogDigitizer installer for Windows and follow the installation instructions. Once installed, open the application from the Start menu or the desktop shortcut. You do not need to install Python separately.

Before starting, choose a scanned log with clearly readable depth references and curve-scale values. Calibration requires at least two known depth references and two known horizontal values for each curve scale.

1.2. Create a project and add images

A project keeps the imported images, calibrations and digitized curves together.

1. Select **File → New Project**, enter a project name and choose where to save it.
2. Confirm the new project. The image-selection dialog opens automatically.
3. Select the scanned image or images you want to digitize.

The imported images appear in the **Project Explorer** on the left. Select an image to display it in the viewer. You can add further images at any time using **Add Image**.

A project may contain consecutive images covering different depth intervals, or images covering the same interval but containing different curves. Each image is prepared and calibrated individually; curves can then be combined during export.

Use **Save Project** or **Ctrl+S** regularly. To continue an earlier session, open its `.logproj` file or select it from **Recent Projects**.

Moving a project: Keep the `.logproj` file and its associated project folders together when copying the project to another location or computer.

1.3. Find your way around

The workspace has three main areas:

- **Project Explorer — left:** select an image, a curve and the task you want to perform.

- **Image viewer — centre:** inspect the scanned log, place calibration references and digitize curves.
- **Task Panel — right:** access the controls for **Preprocessing**, **Calibration** or **Digitization**.

Use the mouse wheel to zoom around the pointer. To move along the image, activate **Pan** and drag with the left mouse button. Press **F** to fit the image to the viewer, or **1** to display it at its original pixel scale.

The floating readout in the viewer shows depth and curve value as you move the pointer over a calibrated image. These values describe the pointer position. Over a saved digitized section, the readout uses that section's scale and offset, and warns if they differ from the active settings in the Task Panel.

Hover over an unfamiliar icon to see its function. The information buttons beside the task controls provide additional guidance.

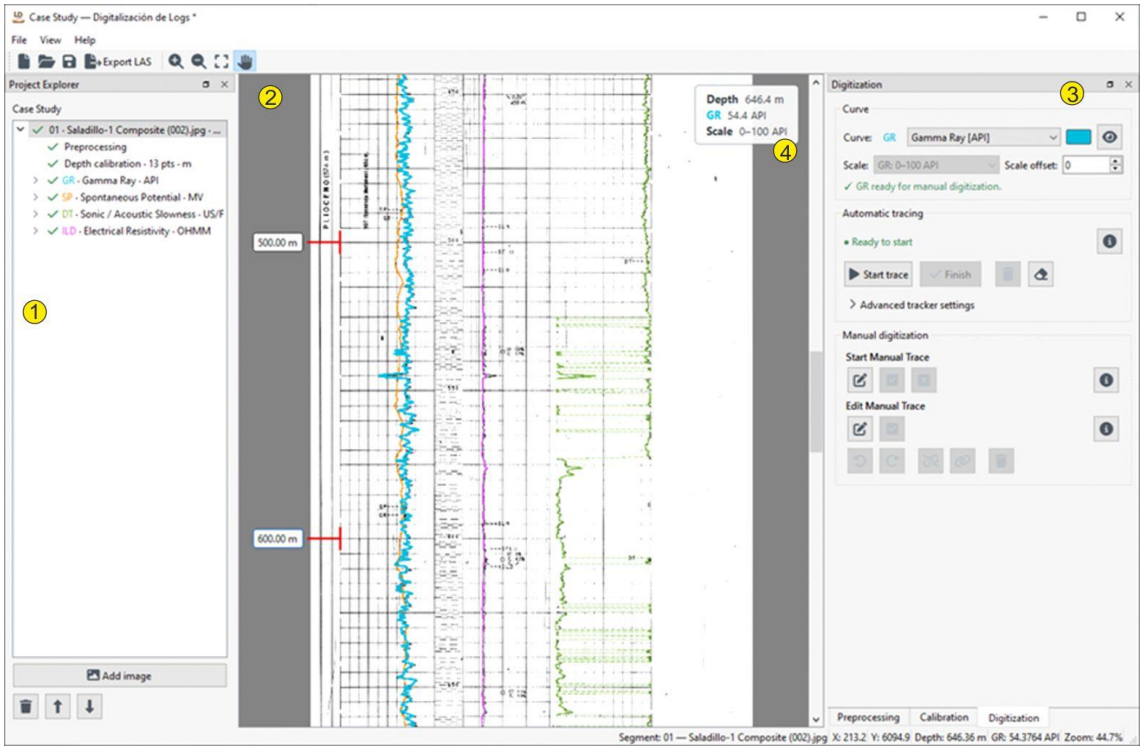


Figure 1.1. The LogDigitizer workspace: (1) Project Explorer, (2) image viewer, (3) Task Panel and (4) floating depth and curve-value readout.

1.4. Follow the workflow

For your first curve, work through these stages:

Stage	What to do
Project & Images	Create the project and add the scanned log images.
Prepare	Rectify the image and adjust its appearance where needed.
Calibrate	Relate image positions to known depths and curve values.

Digitize	Trace the curves semi-automatically and correct difficult intervals manually.
Quality Control	Review curve coverage, gaps and the assembled LAS curves.
Export	Select the curves and sampling interval and save the LAS 2.0 file.

Complete image preparation before calibration: rectification controls become locked once an image contains calibration or digitization data. Image Enhancement remains available and may be revisited during digitization.

2. Prepare and calibrate the image

Before tracing a curve, prepare the scanned image and relate its positions to known depth and curve values. Accurate calibration is essential: a trace may follow the printed curve perfectly and still produce incorrect data if its references or units are wrong.

Select the image in the **Project Explorer**, then open **Preprocessing**.

2.1. Rectify the image

Rectification corrects the geometry of a tilted or distorted scan using four corners around the log area. This step is required before calibration.

1. Activate **Define 4 corners** and place the corners on clear references defining the rectangular log area.
2. Zoom in and adjust their positions where necessary.
3. Open **Preview** and inspect the alignment of the log boundaries and grid.
4. When the result is satisfactory, select **Apply Rectification**.

The corners define the reference geometry; they do not crop the scan. The complete image is retained in the corrected result.

If the lateral boundaries remain curved, use **Multipoint correction** before applying the rectification. Select **Add pair**, then place a point on each boundary at the same transverse reference on the log. Repeat only where additional alignment is needed, and inspect the preview again.

Review the geometry before selecting **Apply Rectification**: the corner and multipoint controls become unavailable after it is applied.

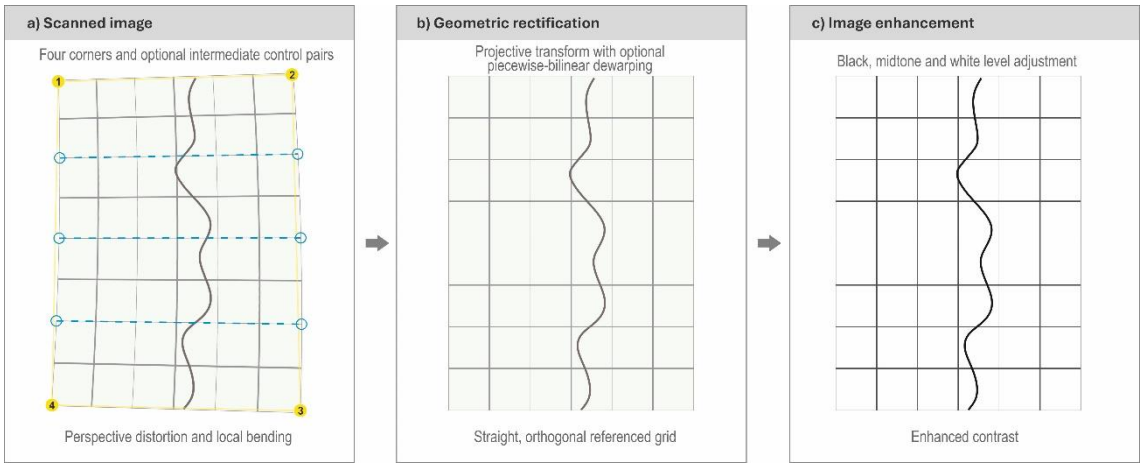


Figure 2.1. Image preprocessing in LogDigitizer. (a) Scanned log with four corner references (yellow) and optional intermediate control pairs (blue). (b) Rectified image after correcting tilt and local distortion. (c) Enhanced image after adjusting black, midtone and white levels to improve curve visibility.

2.2. Improve image clarity

After rectification, use **Image Enhancement** if the curves are faint or the paper background is uneven. Adjust **Black**, **Midtones** and **White** while inspecting the image, then select **Apply Enhancement** to retain the changes.

Aim for a clearly distinguishable curve while preserving thin lines and faint intervals. Strong contrast adjustments can remove useful detail or make the curve harder to separate from the grid.

Enhancement is optional and can be revisited at any time, including during digitization. Different parts of a scanned log may require different adjustments, so you can return to **Image Enhancement** and modify the settings to suit the interval you are working on. These adjustments affect the image as a whole, rather than only the visible area.

2.3. Calibrate depth

Open **Calibration** and use **Depth calibration (Y)** to connect vertical image positions to borehole depth.

1. Select the depth unit, **m** or **ft**, before adding the first point.
2. Activate **Add Y Point**, click a clearly identified depth reference and enter its value.
3. Repeat at a second reference, preferably well separated from the first and spanning the interval you intend to digitize.
4. Add further references where needed to account for variations in vertical spacing, then deactivate the capture tool.

For example, if two labelled grid lines represent 100 m and 150 m, place a calibration point precisely on each line and enter those depths. Use the grid line itself rather than the position of its printed label.

Depth values must increase or decrease consistently along the image. The panel indicates when the calibration is valid.

Check the result against another labelled depth reference, if available, using the floating readout in the viewer. If it does not agree, review the point positions, entered values and selected unit.

2.4. Define the curve and calibrate its scale

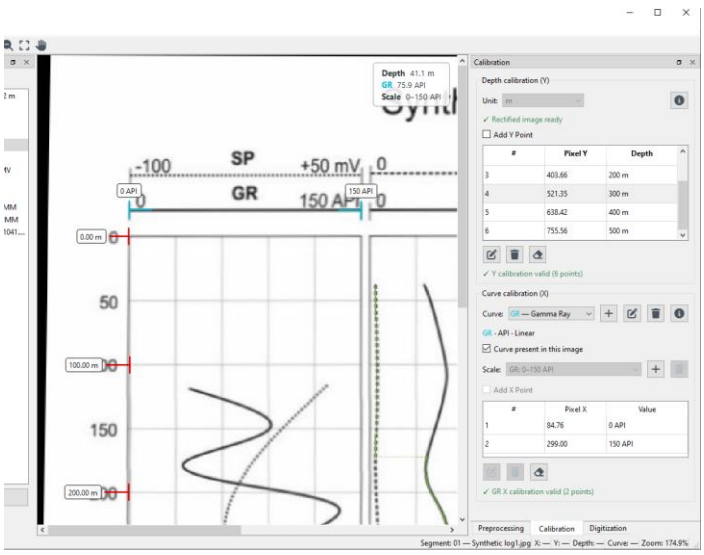
In **Curve calibration (X)**, create or select the curve you want to digitize. When creating a curve, choose a predefined parameter or enter its details. Check the mnemonic, unit and whether the printed scale is **Linear** or **Logarithmic**.

For this first example, consider a gamma-ray curve identified as **GR**, with a linear scale from **0 to 150 API**.

1. Select the curve and ensure **Curve present in this image** is enabled.
2. Activate **Add X Point**, click the reference line corresponding to **0 API** and enter `0`.
3. Click the reference line corresponding to **150 API** and enter `150`.

Use references belonging to that curve's printed scale. The two horizontal positions and their values must differ; logarithmic scales require positive values.

Each curve is defined once for the project, but its horizontal scale must be calibrated separately in every image where it appears. Reuse the existing curve definition when moving to another image.



Additional scales and wrap-around curves are covered in Chapter 4.

Figure 2.2. Depth and curve-scale calibration for a gamma-ray log. Depth markers coincide with labelled horizontal grid lines; X-calibration markers identify the 0 and 150 API references. The Calibration panel shows the selected curve and valid calibrations.

2.5. Check before tracing

Before opening **Digitization**, check that:

- Depth readings agree with the printed references and use the correct unit.
- The selected curve, unit and scale type match the log.
- Curve-value readings agree with the horizontal scale.

The viewer readout reports the value corresponding to the pointer position; you do not need a digitized trace to check calibration.

To refine a reference, drag its marker. To correct an entered value, double-click its label. Vertical position determines depth calibration; horizontal position determines curve-scale calibration.

Once these checks are complete, the image is ready for tracing.

3. Digitize and correct curves

Use semi-automatic tracing to capture the curve, then correct difficult intervals manually. Both methods contribute to the same output curve, so you can combine them as needed.

Select the image and open **Digitization**. Check the active **Curve**, **Scale** and **Scale offset** before starting. For a curve using its normal printed range, leave the offset at **0**.

3.1. Start a semi-automatic trace

In **Automatic tracing**, select **Start trace** and click precisely on the curve near the top of the interval you want to digitize. Choose a clear point away from grid intersections or annotations.

The tracker follows the curve downwards through the image and produces a temporary draft. Inspect it as you move along the log, checking that it follows the intended curve rather than a grid line or a neighbouring trace.

The draft remains available for correction until you select **Finish**.

3.2. Correct or resume the draft

If the tracker leaves the curve, click the correct curve at the depth where you want tracking to restart. The draft above that depth is retained, and the lower part is replaced by a new trace from your selected point.

Choose the correction point near the beginning of the deviation so that an incorrect interval is not left above it.

If tracking stops at an interruption, click the curve where it becomes clear again to resume. Inspect the connection across the interruption; use a manual section later if it needs a more accurate reconstruction.

Repeat these corrections as necessary, then select **Finish** to retain the trace as a saved section. Use the draft's trash icon if you want to discard it and start again.

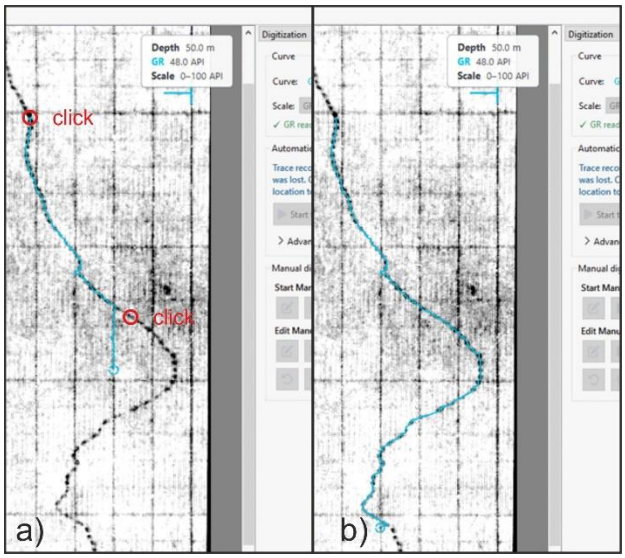


Figure 3.1. Correcting a semi-automatic trace. (a) The tracker leaves the curve and follows a grid line. A new point is selected on the correct curve near the start of the deviation. (b) Tracking resumes from that point, preserving the upper part of the draft.

3.3. Adjust tracking when needed

Start with the default settings. If the same problem occurs repeatedly, open **Advanced tracker settings** and adjust the control most relevant to it.

Problem	Adjustment to try
A faint curve is not detected reliably	Increase **Sensitivity**.
The trace jumps to nearby lines	Reduce **Search width**.
Abrupt lateral changes are missed	Increase **Search width**.
Sharp bends are not followed	Increase **Flexibility**.
Short breaks repeatedly stop tracking	Increase **Gap tolerance**.

Change one setting at a time and review the result. Higher values are not always better: a wider search or greater gap tolerance can also connect unrelated features.

Finish or discard the current draft before changing tracker settings. If poor image contrast is the main difficulty, return to **Image Enhancement**, adjust the image and apply the changes before trying again.

For a short, ambiguous interval, manual correction is often quicker than repeated adjustments.

3.4. Complete difficult intervals manually

Use **Manual digitization** where the curve crosses another line, becomes faint or requires a more carefully controlled reconstruction. You can also digitize an entire curve manually.

1. Check the curve, scale and offset, then select **Start Manual Trace**.
2. Place nodes along the printed curve, progressing consistently in one vertical direction.
3. Inspect the curve drawn between the nodes and select **Finish Trace** when it follows the intended path.

At least two nodes are required. Use fewer nodes along smooth intervals and add more around bends or rapid changes.

To correct a saved automatic trace, start the manual section just before the error and end it just after, placing both endpoints on reliable parts of the curve. **The manual section replaces the automatic result within its depth interval**, while the automatic data outside it are preserved. You do not need to erase the automatic interval first.

Use the same scale and offset as the interval being corrected. The floating readout helps identify the settings assigned to the saved section.



Figure 3.2. Manual correction of an automatic result. Left: an incorrect trace at a curve intersection. Right: control nodes define a replacement section that follows the intended curve and connects with the reliable automatic intervals above and below.

3.5. Refine saved sections

To refine a manual section, activate **Edit Trace** and select it in the viewer. Its control nodes become visible:

- Drag a node to adjust the curve.
- Click between existing nodes to insert another.
- Right-click a node to remove it.

Select **Done Editing** when finished. Automatic sections do not expose editable nodes; correct them with a manual replacement or retrace the affected interval.

To remove an unwanted part of a saved automatic trace, activate **Erase automatic range**, start dragging directly on the trace and drag vertically across the interval. Release to apply the removal. Manual corrections within that range are preserved.

Use **Undo** (**Ctrl+Z**) or **Redo** (**Ctrl+Shift+Z**) to revise digitization edits. This history applies to the current image and is cleared when you switch images; discarding an unfinished draft cannot be undone.

3.6. Review the completed curve

Inspect the full digitized interval before moving on. Pay particular attention to curve intersections, sharp peaks, faint sections and the connections between automatic and manual sections.

Check that the trace follows the intended curve, that no unwanted gaps remain and that each section uses the correct scale and offset. Where the original curve cannot be interpreted reliably, leave a gap rather than inventing a continuation.

Finish any active draft and save the project. You can then select another curve or image, or proceed to export.

The next chapter explains how to work with additional scales, wrap-around curves and multiple images.

4. Work with scales and multiple images

Some logs use more than one horizontal scale or continue across several scanned images. Before digitizing these cases, identify how the curve is represented: a change in its printed range requires an additional scale, while a curve that wraps around the track boundary requires a scale offset.

4.1. Use an additional scale

A curve may use two printed ranges, such as **GR 0–100 API** and **GR 0–1000 API**. Define these as separate scales belonging to the same curve.

In **Calibration**, select the curve and complete its first X calibration. Then use **Add Scale** and place two X-calibration points for the additional range. Each scale is named automatically from its calibrated values and unit.

In **Digitization**, select the scale corresponding to the interval you are tracing. When the printed range changes:

1. Finish the current section.
2. Select the appropriate **Scale**.
3. Start a new section using that scale.

Each saved section retains its assigned scale. Selecting another scale for subsequent tracing does not change previously saved sections.

4.2. Digitize a wrap-around curve

On some analogue logs, the recording pen reaches one edge of the track and continues from the opposite edge. The apparent jump represents a continuation beyond the printed range.

Use **Scale offset** to recover the actual values. For a linear scale from **0 to 100 API**:

Scale offset	A position reading 30 API on the printed scale represents…
0	30 API
+1	130 API
+2	230 API
−1	−70 API

Choose the offset from the log's recording convention and the continuity of the curve.

At a wrap-around transition, finish the current section, change **Scale offset** and start another section on the opposite side of the track. Return the offset to **0** when the curve returns to its base range.

Do not trace directly across the horizontal jump. Separate sections allow each part to retain the correct offset. Scale offsets are available for linear scales.

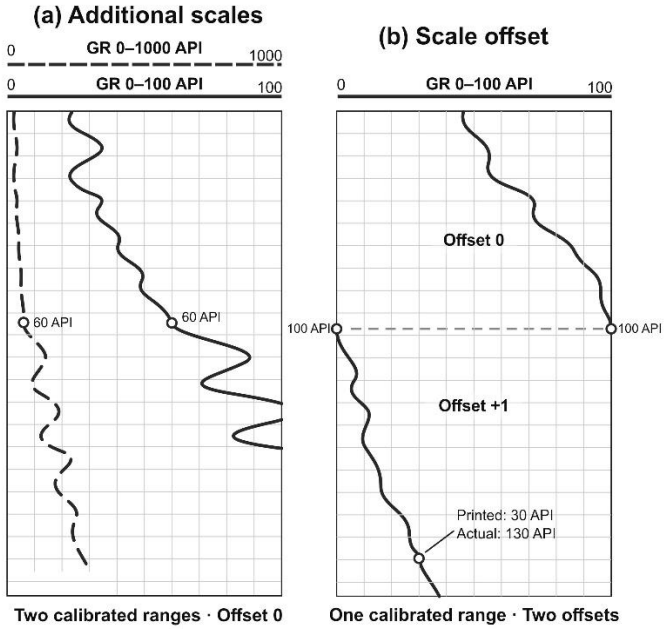


Figure 4.1. Additional scales and scale offsets. (a) A curve uses two independently calibrated printed ranges. (b) A curve wraps around a single printed range; separate sections with offsets of 0 and +1 preserve its numerical continuity.

4.3. Check a saved section's scale and offset

Move the pointer over a saved section and inspect the floating readout. It uses the scale and offset assigned to that section and warns when they differ from the active settings in the Task Panel.

Check this context before adding a manual correction. For example, a section saved with offset **+1** still requires that offset even if the panel currently shows **0**.

For an editable saved section, use **Edit Trace**, select the section, adjust **Scale** or **Scale offset**, then select **Done Editing**. Review the resulting values before continuing.

4.4. Continue a curve across consecutive images

When a long log has been scanned in parts, add the images to the same project and arrange them in their intended order in the **Project Explorer**.

Prepare and calibrate each image independently. Reuse the same project curve wherever it continues, and keep **Curve present in this image** enabled for those images.

For overlapping images:

1. Calibrate depth and the curve scale in both images.
2. Select the curve and locate the red dashed overlap guide.
3. Extend digitization slightly above or below the guide, following its displayed instruction.
4. Finish the trace before switching to the neighbouring image, then follow its corresponding guide.

The guides identify where coverage should extend so that the curve can continue between images. They are references for digitization, not cut lines or editable points.

Overlap guides are calculated **for the selected curve**. An image participates only when that curve is marked as present, so different curves may have different neighbouring images.

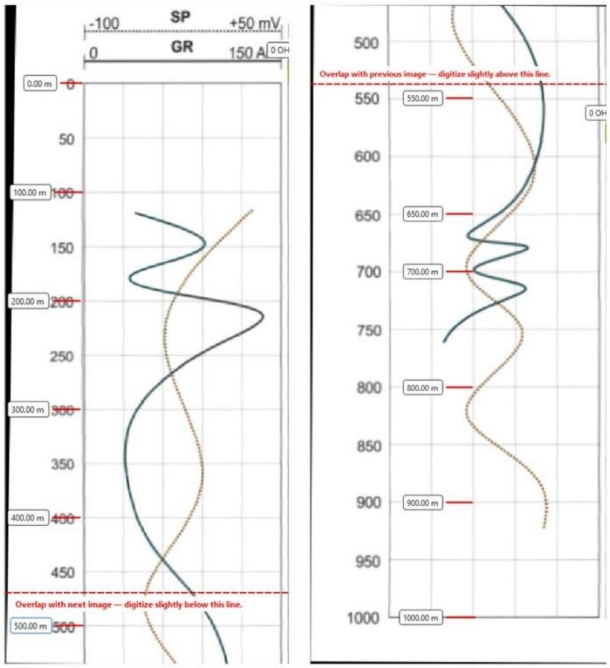


Figure 4.2. Continuing a curve across overlapping images. The red dashed guides indicate where digitization should extend in each image. Following both instructions provides coverage across the transition.

4.5. Combine images containing different curves

Images do not have to form a single vertical sequence. For example, one set may contain **GR** and **SP**, while another contains **DT** over the same depth interval. These can all belong to one project and be exported together.

Create each curve once, then indicate where it appears using **Curve present in this image**. Calibrate depth in every image and calibrate the horizontal scales only for the curves present there.

For example:

Image	Curves marked as present
Image A	GR and SP
Image B	DT

Image B does not affect the overlap guides for GR or SP. The same curve-specific approach is used when checking gaps for LAS export.

Before exporting, review the coverage of each curve separately. A complete depth interval for one curve does not mean that the others contain data throughout that interval.

5. Quality control

Before exporting the digitized curves, use **Quality Control** to check their sampled coverage and identify intervals that may require attention. Quality control helps detect small gaps, incomplete intervals and other discontinuities that may not be obvious when reviewing the trace directly on the scanned image.

Open **Quality Control** from the toolbar or from **File → Quality Control**. The window summarizes the digitized curves and evaluates their coverage using the selected depth sampling interval (**STEP**).

Quality Control does not modify the original digitized sections. It provides a final review stage between digitization and LAS export.

5.1. Choose the sampling interval

STEP defines the regular depth interval used to evaluate the digitized curves. It is also the sampling interval that will be used during LAS export.

For example, a STEP of 0.10 m evaluates whether a curve value can be obtained every 0.10 m throughout its digitized interval.

Choose a STEP appropriate to the detail visible in the original log and to the intended use of the exported data. A smaller STEP produces more samples but does not increase the resolution of the scanned source. A very small STEP may also reveal short gaps between digitized sections that would not be detected with a coarser interval.

The selected STEP is shared with **Export LAS**. Changing it in Quality Control therefore changes the sampling interval that will initially appear when the export window is opened.

5.2. Review detected issues

Quality Control evaluates each curve separately and lists intervals where sampled coverage requires attention.

Review the summary for each curve before export. Reported intervals may correspond to:

- a short gap between digitized sections;
- incomplete digitization;
- a transition between images;
- an interval without interpretable data in the original log; or
- a small sampling gap that can be interpolated from neighbouring values.

The issue status indicates how the interval will be treated. Intervals that can be safely reconstructed may be identified as **interpolated**, while other intervals may require **review** or may be deliberately **ignored** when the absence of data is intentional.

Do not correct a reported interval automatically without checking the scanned record. A gap may represent unfinished digitization, but it may also correspond to genuinely missing or unreadable source data.

Where the original curve is visible, return to the corresponding image and complete or correct the digitization. Where the source information is absent or cannot be interpreted reliably, retain the gap rather than creating an unsupported continuation.

Eligible short internal gaps are interpolated only in the sampled LAS output; the original digitized sections remain unchanged.

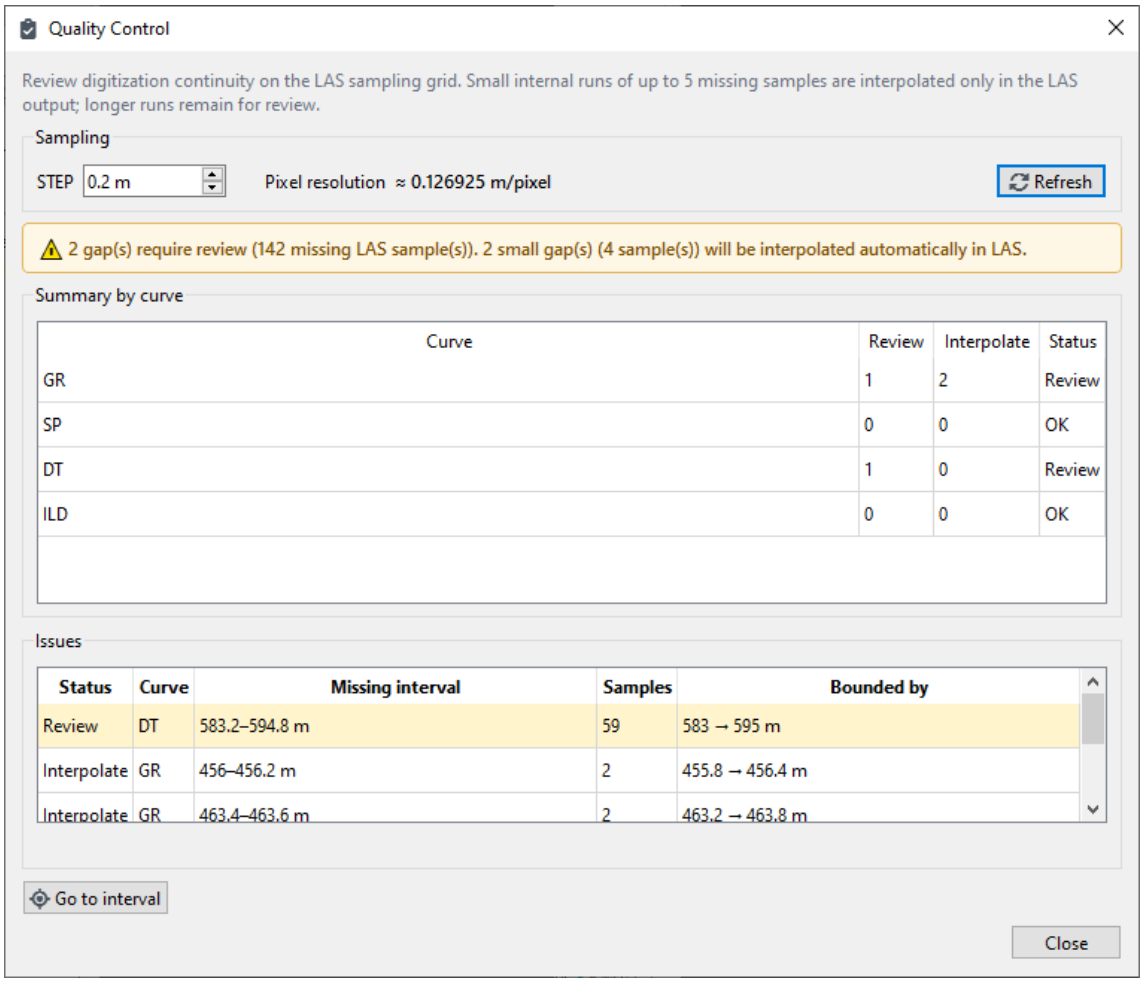


Figure 5.1. Quality Control window. Digitized curves are evaluated using the selected STEP. The summary identifies intervals that have been interpolated, ignored or require review before LAS export.

5.3. Inspect an issue in the original image

Select a reported interval and use **Go to interval** to inspect it in the original scanned log.

LogDigitizer opens the corresponding image, centres the viewer on the affected depth and adjusts the view so that the interval can be examined together with the surrounding curve.

Compare the digitized trace with the source image before deciding how to resolve the issue. Pay particular attention to:

- transitions between automatic and manual sections;
- curve intersections;
- faint or interrupted printed curves;
- sharp peaks;
- wrap-around transitions; and
- boundaries between consecutive images.

If the problem results from an incomplete or incorrect trace, correct the digitization and reopen Quality Control to verify the interval.

A reviewed issue does not necessarily have to be removed. When the original log contains no reliable information, a missing interval is preferable to an artificial reconstruction.

5.4. Preview the LAS curves

Use **LAS Curve Preview** to inspect the digitized curves in the form in which they will be sampled for LAS export.

The preview combines the digitized sections belonging to each curve across the project and displays them against depth using the current STEP. This provides a convenient way to review the overall result without having to inspect each scanned image separately.

Use the preview to check:

- the general shape and continuity of each curve;
- transitions between automatic and manual sections;
- joins between consecutive images;
- additional-scale transitions;
- wrap-around intervals using scale offsets;
- the location of missing data; and
- whether distinctive peaks and changes occur at the expected depths.

The preview is particularly useful for detecting problems that are difficult to recognise when viewing individual digitized sections in the image viewer.

LAS Curve Preview does not modify the digitized data and does not create a LAS file. If a problem is found, return to the corresponding image, correct the calibration or digitization, and review the curve again.

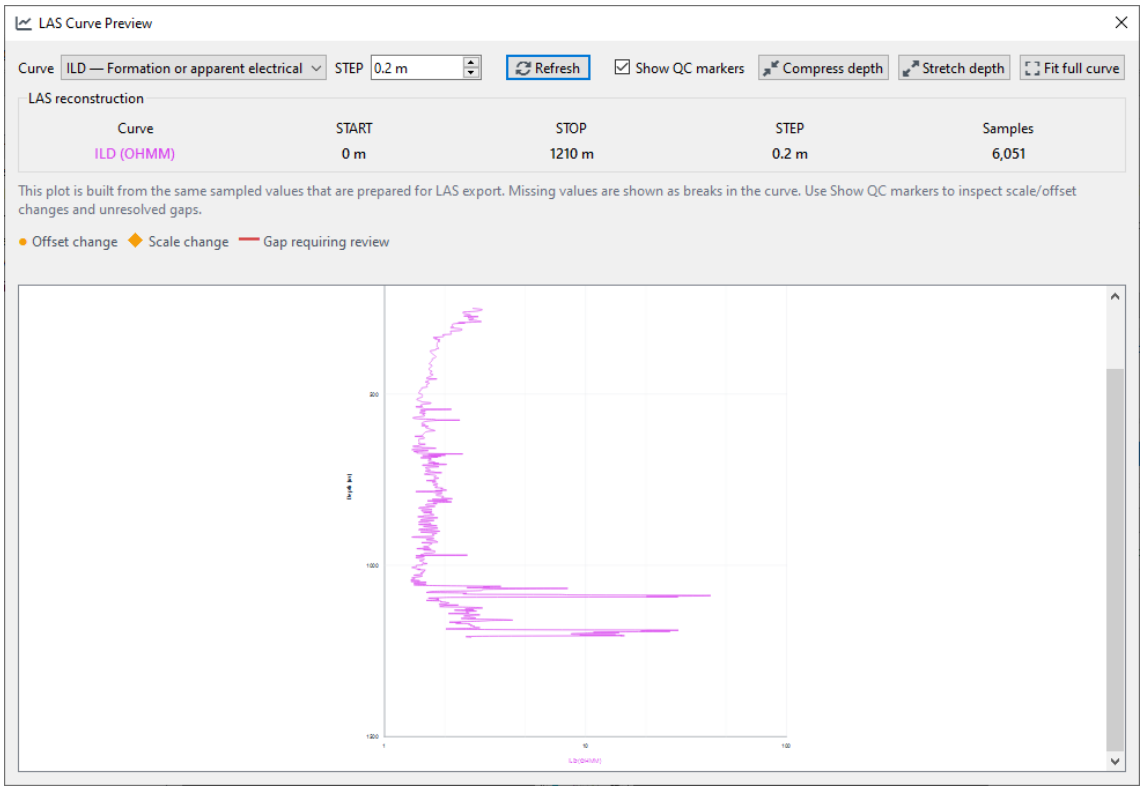


Figure 5.2. LAS Curve Preview. Digitized curves are displayed against depth using the current LAS sampling interval, allowing continuity, gaps, image transitions and overall curve behaviour to be reviewed before export.

Once the reported issues have been reviewed and the curve preview is satisfactory, proceed to LAS export.

6. Export and check the results

LogDigitizer exports the selected curves to a **LAS 2.0** file for use in well-log interpretation and other geoscientific applications. Sections belonging to the same curve are combined across images according to their calibrated depths.

Complete Quality Control, finish any active trace and save the project before opening **File → Export LAS…**.

6.1. Select the curves

In **Data and sampling**, select the curves you want to include in the LAS file. Check their mnemonics, units and digitized ranges.

Only curves with valid calibration and saved digitized data can be exported. If a curve is unavailable, hover over its row to see the reason, then return to calibration or digitization as needed.

Curves included in the same LAS file do not need to cover identical depth intervals. Missing values in one curve do not prevent other curves from containing valid data at the same depth.

All selected curves must use the same depth unit. Images calibrated in metres and feet cannot contribute to the same LAS export.

6.2. Review the sampling interval

STEP defines the regular depth interval between successive samples in the exported file.

The value initially shown is the STEP used in Quality Control. Review it before export and change it if necessary.

For example, a STEP of 0.10 m produces one sample every ten centimetres.

Choose an interval appropriate to the detail visible in the original log and the requirements of subsequent analysis. A smaller STEP produces more samples but does not increase the accuracy or resolution of the source image. An excessively large STEP may miss narrow features.

The export dialog also displays the approximate **Pixel resolution**, expressed as depth per image pixel. Use this as a reference when selecting STEP. Sampling at a finer interval is possible, but the additional samples represent interpolation rather than new information from the scanned log.

LAS START and **LAS STOP** are calculated automatically from the selected curves and STEP. Review these values together with the estimated number of rows before continuing.

Missing samples are written using the LAS null value, **−999.25**. This represents absent data and must not be interpreted as a measured curve value.

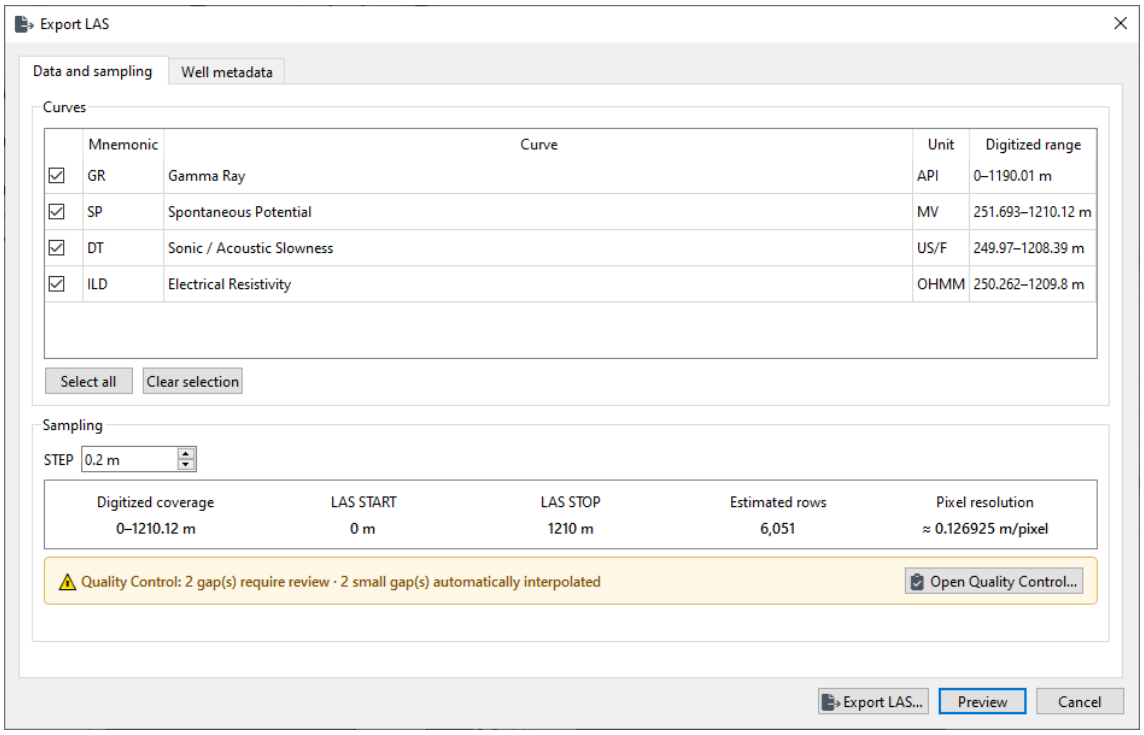


Figure 6.1. LAS export settings. Select the curves, review the depth sampling interval (STEP), and check the output depth range, estimated row count and approximate pixel resolution.

6.3. Add well information

Open **Well metadata** and review the available borehole identification information.

Enter information from the original log where possible and leave unknown fields blank. Check the borehole name and any other available identifiers carefully, because this information will be written to the LAS file.

Use **Other information** for relevant notes about the source record, digitization or any limitations that should accompany the exported data.

Well metadata does not affect the digitized curve values, but complete and accurate identification makes the resulting LAS file easier to interpret and reuse.

6.4. Preview the LAS text and export

Before creating the file, select **Preview** to inspect the LAS text generated from the current export settings.

This preview shows the LAS structure, including:

- well information;
- curve definitions and units;
- START, STOP and STEP;

- the null value; and
- the sampled curve data.

This is different from **LAS Curve Preview** in Quality Control. LAS Curve Preview provides a graphical check of the curves, whereas the export preview shows the actual text structure and sampled values that will be written to the LAS file.

The text preview is read-only. It does not save a file or modify the project. Close it to make any necessary changes to the export settings or metadata.

Review the export summary and resolve any blocking errors. When the settings are satisfactory, select **Export LAS…** and choose the filename and location.

The STEP and metadata used for a successful export are retained in the project. Save the project again to preserve these settings for the next session.

6.5. Check the exported file

Open the exported LAS file in your usual well-log application and plot the curves against depth.

Check that:

- the expected curves and units are present;
- the depth range and sampling interval are correct;
- distinctive peaks and changes occur at the expected depths;
- additional-scale transitions are correct;
- wrap-around sections have plausible continuity;
- joins between consecutive images are consistent; and
- missing intervals are recognised as null data.

Compare several distinctive features with the scanned log, especially near manually corrected intervals, image boundaries and any intervals previously identified during Quality Control.

If you find an error, return to the LogDigitizer project, correct the calibration or digitized section, repeat Quality Control and export the LAS file again.

Keep the LogDigitizer project together with the exported LAS file so that the digitization can be reviewed, corrected and reproduced later.

7. Licence and citation

7.1. Software licence

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7.3. Citation

If LogDigitizer is used in scientific work, please cite the specific software version used.

Citation metadata are provided in the CITATION.cff file and in the LogDigitizer repository:

https://github.com/CarlosMarinLechado/LogDigitizer

For LogDigitizer v1.1.0, please cite:

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